GlucoTrack

Diabetes Management Platform

Business & Enterprise Analysis

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Section 1: Business and Enterprise Analysis

Research

1. Industry segment

Glucose Tracker operates in the digital healthcare industry. We focus on diabetes management through a mobile app that monitors your glucose levels.

2. Products and services

The main product is a mobile app called Glucose Tracker. This app will feature:

- Real-time glucose monitoring
- Graphs that show trends and weekly reports
- Alerts for high and low blood sugar
- Predicts high and low blood sugar

3. Market

Glucose Tracker is designed for individuals with type 1 diabetes and insulin-dependent type 2 diabetes who rely on continuous glucose monitoring (CGM) devices. Diabetes has been a growing health issue, and the demand for smart health tools has increased significantly.

Glucose Tracker positions itself within this market as a software solution that improves existing CGM systems by offering clearer trend analysis, predictive alerts, and improved usability.

4. Business and Organizational Structure

We will develop an app that includes basic glucose tracking for free while offering a premium subscription for:

- Predictive analytics
- Advanced reports
- Provider data-sharing tools

Glucose Tracker will also have partnerships with healthcare providers and possible licensing partnerships with CGM manufacturers.

Value Chain

1. Primary Activities

Inbound Logistics

Developing secure API Connectors for various CGM devices such as Dexcom, Abbot (Libre), and Medtronic. As well as manual entry for users without these high-end sensors.

Operations

Running raw glucose data through ML models to predict low “Hypo” or high “Hyper” events before they occur. Every 30 minutes before they happen.

Outbound Logistics

Generating Automated AGP (Ambulatory Glucose Profile) reports. Push notification systems via mobile/app that can alert the user along with caregivers or guardians (followers) if a dangerous condition level is reached.

Marketing & Sales

Target Endocrinology clinics rather than just individual users. Using a “B2B2C” model where doctors prescribe your app to patients.

> **Key Definitions**
>
> Endocrinology clinics specialize in diagnosing and treating diseases related to the endocrine system, which regulates hormones, metabolism, growth, and reproduction. They manage chronic conditions such as diabetes, thyroid disorders, and pituitary gland disorders.
>
> B2B2C – Business-to-Business-to-Consumer model where a business partners with another business to access their customers/consumers.

Service

24/7 technical support for syncing issues (critical for health applications). Regular security patches for the mobile app.

2. Support Activities

Firm Infrastructure

- Compliance & Legal: Will need ongoing FDA Class II medical software monitoring and HIPAA audits.
- Financial Management: Managing the subscription (SaaS) revenue model and potential insurance reimbursement codes.

Software as a Service Model (SaaS): a cloud-based software delivery model where applications are hosted by a vendor and accessed by users over the internet. GlucoTrack could pay for the service monthly with a “pay-as-you-go" structure.

HR

- Hiring: Recruiting developers with experience in Bioinformatics or health-technology.
- Medical Advisory Board: Hiring part-time Endocrinologists to validate the predictive algorithms, so the app is clinically reliable.

Technology

- UI/UX Research: Due to the medical information and uses of the project, it is important to focus on the cognitive load that the users will experience. A diabetic user might often be stressed, so the app must be simple to use, especially during a low blood sugar episode.
- AI refinement/training: Constantly training the models that will be used is vital; using anonymized data to improve prediction accuracy.

Procurement

- Cloud Infrastructure: Need to source high-security, encrypted cloud hosting such as AWS, Azure Health, to guarantee data residency and security.
- Third-Party Data: Licensing food nutrition databases to power the Food tracker/Meal logging feature. Help users not have to manually calculate the number of carbs in food which they need to give the right units of insulin which will reduce decision fatigue for users.

Stated Areas of Differentiation

Predictive Clinical Intelligence

In addition to the reports on current and past glucose levels that standard GCM apps provide, GlucoTrack will utilize ML models to provide a forward-looking "window" into the user's health.

- Early notification system predicts the user's hypo/hyperglycemic events 30 minutes before they occur so the user can have the ability to be more proactive in preventing accidents.
- Accuracy is vital, so committing to refining AI through anonymized data sets ensures that GlucoTrack's predictive accuracy improves over time, which will give us an advantage over new entrants as it is difficult to replicate.

Reduced Cognitive Load & User-Centered Design

Managing chronic conditions is difficult and mentally exhausting. GlucoTrack will be made to minimize "decision fatigue", particularly during high stress events.

- With Automated Nutritional Integration through the procurement of 3rd-party nutritional databases will remove the burden of manual carb counting. This will directly improve the accuracy of insulin dose calculations, which is a critical pain point for people with Type-1 diabetes
- Responsive UX: UI/UX research will focus on accessibility and safety during "hypo" events, ensuring that the interface is simple and easy to navigate even in the case when the user is experiencing cognitive impairment due to low blood sugar.

Integrated Care System

- Clinical Endorsement: Targeting endocrinology clinics directly helps us gain trust from clinicians. Also, a Medical Advisory Board ensures our features align with professional medical standards, making the app more than just a lifestyle "utility".
- Factors of Safety: The real-time analytics that display the user's blood glucose levels assist users in understanding their levels, and with the notification system, it makes it a safety net for the user and their followers (guardians/family).

Hardware Agnosticism

Compatibility with multiple hardware brands is possible through developing secure API connectors and offering manual entry fallbacks. This will allow for a single "source of truth" for users who may switch their CGM hardware but also preserve their history and predictive insights.

Analysis

Threat of New Entrants: Moderate

- Regulation requirements are a barrier of entry
    - Must ensure compliance with the FDA
    - HIPAA compliance
    - Must have secure and encrypted data storage and transmission
    - Must have secure authentication and role-based access
    - Involving some level of legal consultation
- Development costs are relatively low as technical tools are becoming more accessible
- Often additional government funding given for health tools (lowers cost)
- Patients often use trusted names when their health is involved, so new entrants will need high levels of clinical approval
- Requires approval from CGM manufacturers

Bargaining Power of Suppliers: High

- If CGM manufacturers deny partnerships with their product, the app will not have access to data and will therefore not function
- Many CGM companies like Dexcom already have their own app for glucose monitoring
- App stores like Google and Apple have decision power on application distribution, users can not download the app without their approval

Bargaining Power of Buyers: High

- Since many CGM devices already come with their own app, there will need to be significant differences and advantages to the startup's app
- If data is inaccurate, users will likely switch apps
- If price is too high, users will have incentive to switch

Threat of Substitute Products or Services: Low

- One alternative is manual logging data, which is tedious and unreasonable in our current age of technology
- Many smartwatches have glucose tracking technology, but it is unreliable and not recommended

Rivalry Among Existing Competitors: High

- CGM manufacturers apps: large incentive for consumers as the app is specifically designed for their device
- Other glucose monitoring startups: not as big of a threat because they are in the same position, but if their app has many advantages, consumers will gravitate towards them

Strategy

The company should aim to create a more technologically advanced application and interface so that it stands out over the existing CGM manufacturer's apps. Potential features include more advanced analytics, AI based glucose monitoring and predictions, personalized insights, and enhanced user experience and ease of use.

Partnerships with healthcare providers to enhance patients' medical experience. To mitigate the high bargaining power of suppliers (CGM manufacturers), GlucoTrack will maintain a robust, multi-vendor API strategy. By ensuring our architecture is compatible with all major players and independent data aggregators (like Glooko or Tidepool), we prevent vendor lock-in and ensure service continuity even if a single partnership is terminated.

Strategy Justification

This strategy is justified because there aren't many other fields that a startup health monitoring app can use to compete with existing companies. For example, competing on cost would be useless as most other glucose monitoring apps are free to use. In addition, the startup will not be successful with just basic features, as there are many well-known brand names that are already established and available. However, this strategy will only work if the features implemented are functioning and if the data and analytics are accurate.

Portfolio of Software Applications

To effectively implement the features of the glucose level tracker which aligns with product vision, certain software applications should be developed. These applications will make the product stand out, offering functionalities which users may find easy to use and highly beneficial, alongside differentiating the product from competitors.

1. Real-Time Glucose Monitoring and Overview

Via Bluetooth, the application connects to a CGM device which transmits glucose data and acts as a monitor for the user's glucose levels. The monitor runs continuously both in-app and in the background, as to always allow for data capture. The user can track their current glucose levels and how their glucose levels fluctuate throughout the day. An alert will be sent out and displayed if glucose levels are too high or low, so a user may react accordingly to correct it.

2. Analytics-Based Meal Finding Service

While using the glucose monitoring app, a service could be included which aggregates the meal-tracking habits of users across restaurants, storefronts, etc., so that users can quickly and easily determine meal locations which will have options for managing their glucose levels based on where they are. This would ensure that if a user with Type-1 diabetes didn't have the necessary supplies to manage their glucose levels, they could quickly find alternatives within a specified radius.

3. Health and Fitness Integration

The glucose monitoring app could optionally allow users to link other health and fitness apps or devices to track how that data coincides with their glucose levels. Analysis of how a user's health and fitness habits correlate with glucose levels could give users insight on how certain habits correlate with their glucose levels. Suggestions based on the data can be provided to let users know what habits are good, and which ones are negative.

4. Emergency Services Support

This application leverages the Push Notification Systems defined in our Outbound Logistics. In the event of a "Critical Low" prediction from our ML models, the system triggers an escalating series of high-priority push notifications.

If the user does not acknowledge the alert within a set timeframe, the Outbound Logistics system automatically escalates the notification to a server-side trigger. This trigger sends a secure, HIPAA-compliant data packet that contains the user's location and last-known glucose levels directly to designated emergency contacts and local emergency services.

This turns our existing communication infrastructure into a life-saving automation, proving the value of our specialized logistics.

5. GlucoBridge Interoperability Layer

To provide a truly hardware-agnostic experience, GlucoBridge acts as a middleware that standardizes data from different CGM brands. Whether a user switches sensors or uses multiple devices, GlucoBridge "normalizes" the incoming raw data into a single unified stream. This ensures that the user's history trends and ML-driven predictions remain uninterrupted during a hardware switch. It transforms the app into a single reliable source of information that outlasts any single sensor's 10–15-day lifespan.

Application to Develop Further: The Predictive Alert & Interoperability System

The Predictive Alert System is the primary vehicle for achieving GlucoTrack's strategy of Clinical Intelligence and Safety Differentiation. This application transforms raw CGM data into proactive medical interventions by leveraging two critical sub-modules: Multimodal Health Integration and the Emergency Support Loop.

The GlucoBridge Interoperability Layer (The Data Foundation)

Instead of analyzing glucose in a vacuum or relying on a single hardware manufacturer, the system utilizes GlucoBridge as a middleware connector.

- Logic: Machine learning models require clean, standardized data to function accurately. GlucoBridge strips away hardware-specific formatting from devices (like Dexcom or Libre) to provide a continuous, normalized stream of GlucoseReading objects.
- The Benefit: This creates true hardware agnosticism, ensuring that the predictive models have a reliable "single source of truth" even if the user switches CGM brands, while also preventing vendor lock-in.

The Predictive Intelligence Engine (Proactive Analysis)

This module acts as the core mathematical brain of the application, running continuously in the background to analyze the normalized data provided by GlucoBridge.

- Logic: The engine utilizes a three-stage internal logic flow: it calculates the current glucose velocity (rate of change), analyzes the last 180 minutes of historical data, and projects a future trajectory.
- The Benefit: By forecasting if a user will cross a critical threshold (e.g., <= 70 mg/dL) within a 30-minute window, the system shifts diabetes management from a reactive state to a proactive one, giving users the time they need to consume carbohydrates before cognitive impairment begins.

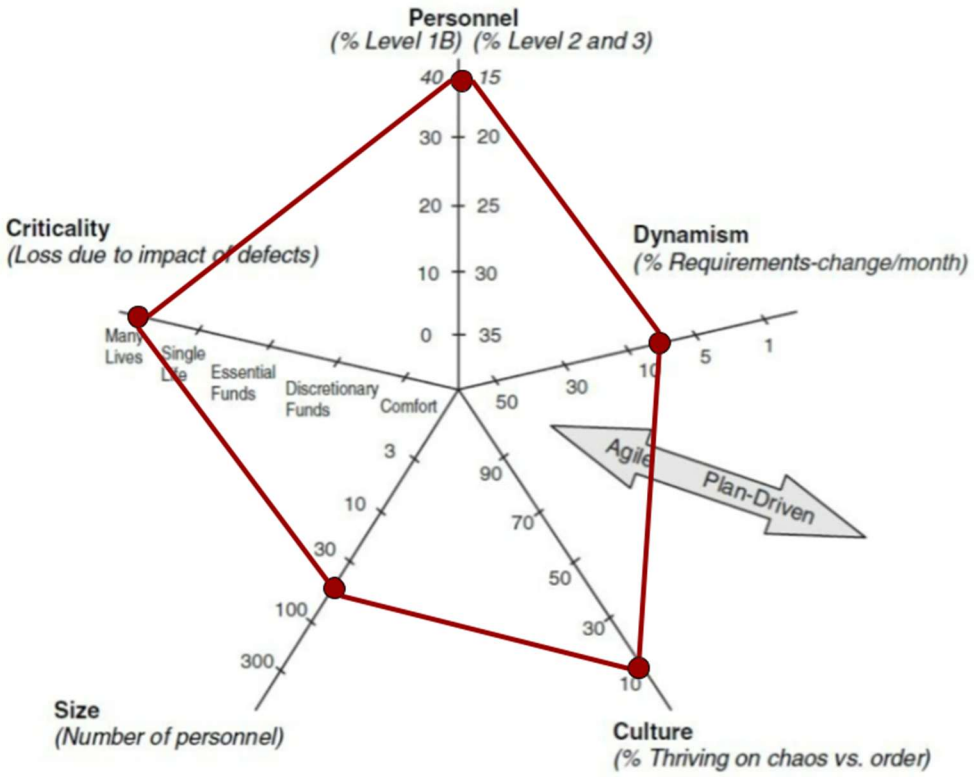
Automated Emergency Services Support (The Fail-Safe)

This module leverages the High-Priority Push Notification System to address the "Critical Response Gap." If the ML models predict a "Critical Low" (< 55 mg/dL), the system initiates an escalating safety protocol:

- The Window: If the user does not acknowledge the high-priority haptic and audible alerts within a 60-second window, the system triggers a server-side escalation.
- The Action: The EscalationManager dispatches a secure, HIPAA-compliant data packet containing the user's precise GPS location and most recent glucose trajectory directly to designated emergency contacts and local EMS providers.
- The Outcome: This turns standard communication infrastructure into a life-saving automation, specifically protecting users during nocturnal hours or periods of cognitive impairment.

Section 2: Intended Software Engineering Process

Spider Diagram Analysis



Personnel

Due to the nature of creating an application for use in a medical context, where users are often making important decisions which will greatly impact their health, it is important to ensure the level of experience within GlucoTrack's personnel base can ensure HIPAA compliance, and capable of ensuring GlucoTrack can properly integrate with medical devices. Therefore, the team will require many high-expertise personnel, who may comprise a compliance team to ensure GlucoTrack remains HIPAA compliant, medical professionals capable of consulting in regard to glucose monitoring and type-1 diabetes, as well as a developer comprised of developers who have prior experience interfacing with vital tracking equipment.

Dynamism

The GlucoTrack application will need to adhere to necessary medical guidelines, meaning that requirements will be less prone to change. However, certain medical guidelines will be updated as new information becomes available, in addition to annual updates to certain guidelines which occur in October. In

Culture

As mentioned previously, a project with applications in a medical setting requires that GlucoTrack's employee base is adhering to strict guidelines and ensuring a proper level of care. The project will also require consulting medical professionals, such as endocrinologists. For these reasons it is unlikely that GlucoTrack's projects would thrive under a culture which is more chaotic. Having orders will ensure consistent communication between developers, project management, and medical professionals.

Size

This project requires both experienced developers and medical professionals to help produce an application with a suite of different features. As such, the number of personnel will likely require somewhere between 40-60 employees, potentially broken into smaller task-oriented teams to implement specific features.

Criticality

GlucoTrack seeks to help those with diabetes who are looking to regulate their glucose levels. Providing inaccurate information or failing to provide information can lead to dangerous fluctuations in an individual with diabetes' health, which can prove fatal in some cases. As such, the criticality of this project is very high, involving many lives.

Process Recommendation

Based on this analysis, a plan-driven approach is recommended, as it will serve the high-criticality, high-precision, and large-scale nature of the project.

Project Management

Release Plan

GlucoTrack operates in a high-critical medical context where an unreliable or inaccurate system can directly endanger users. Given our plan-driven approach and the safety-first nature of healthcare software, our release strategy prioritizes clinical reliability over feature breadth. Each release builds on a verified, stable foundation before introducing additional complexity.

This means Release 1 deliberately excludes lifestyle features such as meal logging and fitness integration — not because they are unimportant, but because they should never be shipped on an untested safety core. A user who relies on GlucoTrack during a hypoglycemic episode at 2 AM cannot afford the emergency escalation system to share infrastructure with an experimental meal-finding feature. The separation of releases reflects this risk management philosophy directly.

Release 1 — Clinical Safety Core (MVP)

Goal: Deliver a clinically safe, HIPAA-compliant glucose monitoring platform that a doctor can prescribe to a patient and trust with their safety.

Release trigger: All Release 1 use cases pass acceptance testing, ML model achieves ≥90% sensitivity on hypoglycemic events, 100% of emergency SMS dispatched within 3 seconds under load.

Use cases/stories it delivers *(Use case/stories point breakdown shown in User Cases & Stories in Section 3):*

- UC1: Check Glucose Data (CGM connectivity via GlucoBridge — Dexcom + Libre)
- UC2: Receive Glucose Alert (real-time high/low threshold notifications)
- UC1-PAS: Generate Proactive Predictive Alert (30-min window)
- UC2-PAS: Escalate Unresponsive Critical Alert (emergency SMS + GPS loop)
- UC4-PAS: Clinical Threshold Steering (endocrinologist adjusts alert sensitivity via Provider Dashboard)
- Story 1: 30-minute predictive alert (college student exam scenario)
- Story 2: Emergency SMS with GPS (caregiver/parent scenario)
- Story 3: Overnight threshold adjustment (endocrinologist scenario)
- NFR: Alert delivery ≤3 seconds, glucose update ≤5 seconds, 0% data loss, MFA, end-to-end encryption

Est. Time: At a sprint velocity of 20 points per 2-week sprint, Release 1 requires approximately **4 sprints (~8 weeks)**

Release 2 — Lifestyle & Multimodal Integration

Goal: Expand GlucoTrack from a safety tool into a daily diabetes management companion by incorporating activity correlation and nutrition.

Release trigger: Health/fitness sync passes interoperability testing with Apple Health and Fitbit. Food logging achieves accurate carb calculation against nutritional database for ≥95% of logged meals.

Use cases/stories it delivers *(Use case/stories point breakdown shown in User Cases & Stories in Section 3)*:

- UC3-PAS: Correlate Activity with Prediction (workout mode, adjusted sensitivity bias)
- Story: Health & fitness correlation (active user adjusting basal insulin)
- UC: Log Meal and Calculate Carbs (food logging with nutritional database API)
- UC: Search Low-Carb Meals by Location (analytics-based meal finding service)
- Story: Automated nutritional logging (user tired of manual carb counting)
- Story: Meal finding near location (Type 1 diabetic without supplies)
- NFR: Fitness sync latency ≤10 seconds, nutritional database coverage ≥500,000 items

Est. Time: At a sprint velocity of 20 points per 2-week sprint, Release 2 requires approximately **3 sprints (~6 weeks)**

Release 3 — Platform Maturity

Goal: Harden the platform for broader clinical adoption — B2B2C scaling, multi-patient provider workflows, and performance under large user load.

Release trigger: Provider Dashboard supports ≥50 concurrent patient profiles per clinic license. Stress test of 5,000 concurrent emergency loop triggers passes.

Use cases/stories it delivers *(Use case/stories point breakdown shown in User Cases & Stories in Section 3)*:

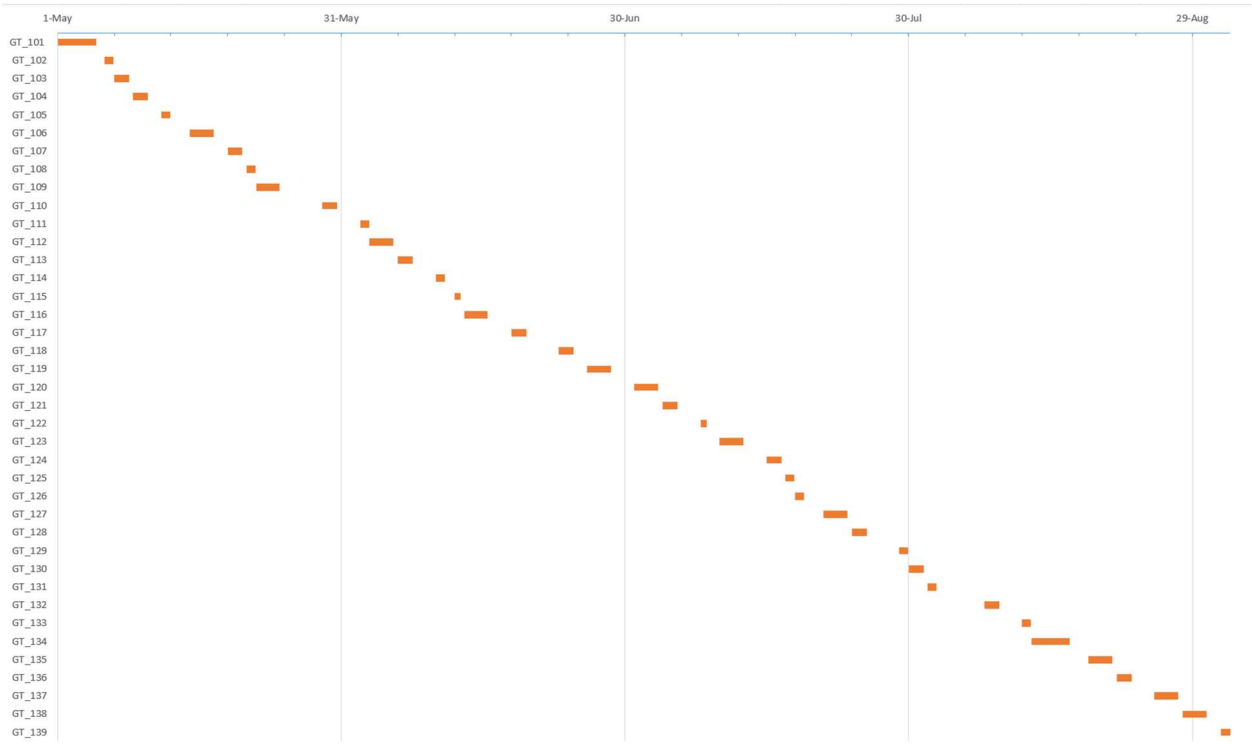
- Enhanced AGP reporting with exportable PDF for insurance/clinical records
- Multi-patient clinic dashboard for endocrinologists
- Performance and load hardening NFRs

Est. Time: At a sprint velocity of 20 points per 2-week sprint, Release 3 requires approximately **2-3 sprints (~5-6 weeks)**

Overall Project Summary

Release	Stories	Pts	25pts/sprint (velocity_	Duration
1	18 stories	97 pts	4 sprints	~8 weeks
2	13 stories	66 pts	3 sprints	~6 weeks
3	8 stories	55 pts	2-3 sprints	~5-6 weeks
Total	**39 stories**	**218 pts**	**~9-10 sprints**	**~19-20 weeks**

Project Schedule



The project schedule shown above spans from the starting date of May 1st, until September 3rd, spanning roughly 20 weeks as speculated in the project summary. This schedule represents the work breakdown for a typical 40-hour 5-day work week, meaning that no stories are started on days that would land on weekends. Each ID listed along the vertical axis represents a user story. Descriptions of these stories can be found in "User Cases & Stories" within section 3 of this document.

Iteration Plan for First Release

Iteration 1: Security and Compliance (Weeks 1-2)

Goal: Establish the HIPPA-compliant backend and encrypted database schemas.

Tasks:

- Research AWS encryption standards and compliant multi-factor authentication protocols.
- Implement end-to-end encryption and multifactor authentication
- Implement role-based access control
- Establish automated backups and a disaster recovery plan, review data residency protocols

Iteration 2: Connectivity & UI (Weeks 3-4)

Goal: Enabling CGM data integration and establishing the primary user interface.

Tasks:

- Analyze Dexcom and Libre API documentation for GlucoBridge
- Develop Bluetooth synchronization layer
- Develop home screen real-time display and interface
- Verify data ingestion latency against 5-second NFR threshold

Iteration 3: Predictive Safety (Weeks 5-6)

Goal: Deploy the prediction engine and emergency fail-safe loops.

Tasks:

- Develop ML prediction models
- Validate engine sensitivity on clinical datasets for 30-minute event windows
- Implement predictive push notifications and unresponsive user GPS escalation loop
- Simulate emergency SMS dispatching under high server load

Iteration 4: Clinical Oversight (Weeks 7-8)

Goal: Finalize dashboard for the clinician steering and AGP reporting.

Tasks:

- Interview endocrinologists on the threshold adjustment workflows
- Build provider dashboard and clinical audit logging for threshold steering
- Conduct final UAT with the Medical Advisory Board

Risk Plan

The development of the GlucoTrack application has several potential risks due to the nature of healthcare technology.

Risk 1: Regulatory Compliance Failure

The application must follow healthcare regulations such as HIPAA. Failure to comply with these regulations could delay development or prevent the application from being released.

Probability: Low

Impact: Extremely High

Transition Indicators:

- Delays in compliance audits
- Failed security or legal reviews
- Changes in healthcare regulations

Risk Management Strategy:
Ensure early and continuous compliance validation throughout development.

Mitigation:

- Conduct regular HIPAA compliance checks
- Consult with legal experts
- Design system with compliance requirements from the start
- Containment
- Delay release until compliance is achieved
- Limit system functionality to non-regulated features if necessary

Risk 2: Data Security Breach

Another important risk involves data security, since the app will store sensitive health information.

Probability: Medium

Impact: High

Transition Indicators:

- Detection of vulnerabilities in security testing
- Unauthorized access attempts
- System alerts indicating abnormal activity

Risk Management Strategy:
Implement strong security measures and continuously monitor system vulnerabilities.

Mitigation:

- Use end-to-end encryption
- Implement secure authentication
- Conduct regular testing

Containment:

- Immediately isolate affected systems
- Notify users and authorities as required
- Patch vulnerabilities and restore secure operations

Risk 3: Prediction Inaccuracy

Prediction accuracy presents a risk because incorrect glucose predictions could lead to unsafe health decisions for users.

Probability: Low

Impact: High

Transition Indicators:

- Low accuracy during model validation
- User reports of incorrect predictions
- Inconsistent prediction performance

Risk Management Strategy:
Continuously validate and improve machine learning models.

Mitigation:

- Train models on large and diverse datasets
- Perform extensive testing before deployment
- Include confidence levels in predictions

Containment:

- Disable predictive features if accuracy falls below threshold
- Revert to basic alert system
- Notify users of potential inaccuracies

Risk 4: Low Clinician Adoption

Low Clinician Adoption is a risk as well for our model; to mitigate this, the Medical Advisory Board (HR Support) will conduct "User Acceptance Testing" specifically for the Provider Dashboard to ensure it fits into a doctor's 15-minute appointment workflow.

Probability: Medium

Impact: Medium

Transition Indicators:

- Low usage of provider dashboard
- Negative feedback from clinicians
- Low partnership engagement

Risk Management Strategy:
Ensure the system aligns with clinical workflows and needs.

Mitigation:

- Conduct user acceptance testing with clinicians
- Incorporate feedback into design iterations
- Involve a medical advisory board

Containment:

- Shift focus toward direct-to-consumer model
- Simplify provider features
- Offer training and onboarding support

Risk 5: Budget Overrun

This project involves high fixed costs due to regulatory compliance, and a significant monthly infrastructure overhead. As a medical platform, unforeseen legal fees or audit requirements could quickly exhaust the $100,000 development budget.

Probability: Medium

Impact: High

Transition Indicators:

- Monthly cloud infrastructure rate exceeding the $30,000 upper estimate
- Legal fees for HIPPA compliance exceeding the allocated budget
- Slower than expected intake of premium users

Risk Management Strategy:
Align infrastructure costs with user demand.

Mitigation:

- Implement pay-as-you-go cloud structure to align with user growth
- Apply for healthcare innovation grants

Containment:

- Shift project to release 1 only to preserve cash
- Defer release 2 lifestyle features until a stable revenue stream is established with clinics

Risk 6: Timeline Exposure

The project schedule estimates a 10-month timeline for Release 1, but the complexity of achieving the ML sensitivity triggers, along with passing regulatory requirements has a risk of leading to delays.

Probability: Medium

Impact: High

Transition Indicators:

- The ML sensitivity remains below the 90% threshold by the end of Iteration 4
- Delays in GlucoBridge API approval from CGM manufacturers
- Regulatory audits taking longer than the allocated 1-month window

Risk Management Strategy:
Develop ML in parallel with the core project.

Mitigation:

- Work on the development of the predictive algorithm alongside the core development to allow for a 2-month buffer for ML tuning
- Maintain a multi-vendor API strategy to bypass single-vendor approval bottlenecks

Containment:

- Utilize feature de-scoping
- If ML prediction engine is delayed, Release 1 can be launched as a real-time safety monitor while the predictive system stays in a beta phase

Linear and Parametric Estimation

Backend development: 5 developers × 5 months

Mobile app development: 5 developers × 5 months

Machine learning development: 5 developers × 4 months

UI/UX design: 5 designers × 3 months

Testing and quality assurance: 5 testers × 3 months

Estimated development timeline: Approximately 10 months for the first full release.

Budget Reserve Calculations

Risk	Likelihood	Impact ($)	Risk Exposure (Likelihood x Impact)
Regulatory Compliance Failure	0.1	50,000	5,000
Data Security Breach	0.3	25,000	7,500
Prediction Inaccuracy	0.1	25,000	2,500
Low Clinician Adoption	0.3	10,000	3,000
Budget Overrun	0.3	25,000	7,500

Total Budget Reserve: Total= $25,500

Following standard risk management practices, the budget reserve for GlucoTrack is calculated using risk exposure (Probability × Impact). The total estimated risk reserve is $25,500.

Schedule Reserve Calculation

The GlucoTrack project has an estimated base schedule of 20 weeks. Based on risk management principles, software project schedules can experience up to 30% uncertainty due to factors such as integration challenges, requirement changes, and technical complexity. However, due to the GlucoTrack software being high risk, high criticality, and heavy on integration, it would be better to place a buffer of 15% instead

Schedule Reserve = 20 x .15 = **3 weeks**

Thus, a 3 week schedule reserve is added to the project timeline, so the total adjusted schedule will be 23 weeks.

Section 3: Requirements Analysis

Storyboard

Title: Joe Receives a Low Glucose Warning

In this storyboard, Joe uses the GlucoTrack mobile application to monitor his glucose levels throughout the day. The system predicts a dangerous drop in glucose levels and GlucoTrack sends an alert to Joe so he can act before experiencing hypoglycemia.

Cast

- Joe: A college student with Type 1 diabetes who relies on a continuous glucose monitor to manage his blood sugar levels.
- GlucoTrack App: The mobile health application that collects tracks glucose data
- Dexcom CGM Device: A wearable sensor that measures Joe's glucose levels and sends data to the GlucoTrack application.

Story

1. Joe has had a long day of studying in the library and he has spent hours preparing for his exams. As a result, Joe forgot to check his glucose levels throughout the day.
2. Because the Dexcom he wears is connected to the GlucoTrack app, the sensor in his CGM device has already sent his readings to the app.
3. The GlucoTrack app has been receiving data from his Dexcom all day, but suddenly the system detects Joe's glucose levels are dropping at an unusual rate. The predictive aspect of the system estimates that Joe has 30 minutes before his glucose level becomes dangerously low.
4. As a result, the GlucoTrack app sends an alert to his phone notifying him of the issue: "Warning: Glucose levels are dropping rapidly. Hypoglycemia predicted in 30 minutes. Suggestion: consume carbohydrates."
5. Joe notices the notification while studying and checks the app, which shows him the analytics that made the prediction. Then he finds a snack to eat and shortly after eating the app shows his glucose levels returning to normal.
6. Thanks to the predictive alert, Joe avoided a dangerous hypoglycemic episode and was able to continue studying safely.

Problem Statement

Patients with diabetes require constant glucose monitoring to ensure they don't experience any health complications. In addition, they need to ensure that the food they consume is not negatively impacting their blood sugar. While there are many existing glucose monitoring systems, they don't provide extensive analytics and can be difficult to use. As a result, we want to create an application to help patients with glucose monitoring that has more detailed analytics such as real time glucose monitoring, meal finding services, collaborations with healthcare providers, emergency services, and integration of health features.

Business Case

Problem: (Data Fatigue**)** Patients are overwhelmed by raw numbers from CGMs without actionable context.

Solution: The GlucoBridge Interoperability Layer

By positioning GlucoTracker as the "central nervous system" for diabetes data (syncing with pumps, CGMs, and fitness trackers), you become the primary interface the user touches 10+ times a day.

Revenue Sources & Customer Tiers

Tier	**Target**	**Features**
Tier 1 - Free (The Safety Core) All Users	Any CGM user regardless of device brand	-GlucoBridge connecticity with all major CGM brands (Dexcom, Libre, etc.) -Real time glucose readings and 24 hr trend graphs -basic high/low threshold alerts & patient/caregiver dashboard with follower notifications
Tier 2 - Premium Individual ($9.99/month)	Type 1 diabetics and insulin-dependent Type 2 users who want proactive management	-30-min predictive alert system (the full PredictionEngine) & critical low emergency escalation with GPS SMS loop - Workout Mode/multimodal fitness integration (Apple Health, Fitbit), AGP report generation for personal use and sharing with doctors -location-based meal finding service -Food logging with nutritional database carb calculation

Tier 3 - B2B2C Clinical License (per clinic, per month)	Endocrinology clinics, not individual doctors	Provider Dashboard with multi-patient monitoring, real-time alert visibility for clinicians (see patient critical lows as they happen). - clinical threshold steering –doctors can adjust alertThresholds remotely -exportable AGP PDF reports formatted for insurance and clinical records -audit log of all threshold adjustments for compliance -patient onboarding tools so doctors can prescribe the app directly

Cost & Benefit Analysis

Category	**Item**	**Estimate**
Cost	Development Costs	$100,000
Cost	Infrastructure Costs	$10,000 – 3$30,000/month
Cost	Regulatory Costs	$100,000
Cost	Operational Costs	$200,000
Benefit	Premium Subscription Revenue	Recurring monthly revenue
Benefit	Provider Dashboard Licensing	B2B recurring revenue
Benefit	Reduced Emergency Incidents	Clinical value + retention
Benefit	Improved Glucose Stability	User outcomes + reviews

Use Cases & User Stories

Release 1 — Clinical Safety Core (MVP)

Use Case 1: Check Glucose Data

Actor: User

Description: User views their current glucose levels and trends.

Steps:

1. User opens the GlucoTrack application.
2. The system retrieves the most recent CGM data.
3. The application displays the current glucose reading and a graph of recent trends.

User Stories for UC1 (Subtotal = 21 pts):

ID #	Description	Story Pts.
GT_101	*"As a user with a Dexcom sensor, I want my data to sync automatically to GlucoTrack so I don't have to switch between multiple medical apps."*	13
GT_102	*"As a user, I want to view my current glucose reading on the home screen so I can check my levels within 2 interactions."*	3
GT_103	*"As a user, I want to see a graph of my glucose trends over the last 24 hours so I can understand how my levels have changed throughout the day."*	5

Use Case 2: Receive Glucose Alert

Actors: System, User

Steps:

1. The system continuously monitors glucose data.
2. The system detects high or low levels of glucose.
3. The system sends a push notification alert to the user.

User Stories for UC2 (Subtotal = 8 pts):

ID #	Description	Story Pts.
GT_104	*"As a user, I want to receive a push notification when my glucose crosses a high or low threshold so I can take corrective action immediately."*	5
GT_105	*"As a user, I want my alerts to be delivered within 3 seconds of detection so I can act before my condition worsens."*	3

UC1 for Predictive Alert System: Generate Proactive Predictive Alert

User Stories for UC1-PAS (Subtotal = 16 pts):

ID #	Description	Story Pts.
GT_106	*"As a college student with Type 1 diabetes, I want the system to notify me 30 minutes before my glucose is predicted to hit a Low threshold, so that I can consume carbohydrates and prevent a hypoglycemic episode during high-stress events like exams."*	8
GT_107	*"As a user, I want the prediction engine to use my last 180 minutes of glucose history so that my alerts are based on my personal trend rather than a single reading."*	5
GT_108	*"As a user, I want my predictive alert to suggest a specific corrective action based on my insulin ratio, so I know exactly how many carbs to consume."*	3

UC2 for Predictive Alert System: Escalate Unresponsive Critical Alert

User Stories for UC2-PAS (Subtotal = 16 pts):

ID #	Description	Story Pts.
GT_109	*"As a parent or caregiver of a diabetic child, I want to receive an emergency SMS with the child's real-time GPS location if they do not acknowledge a Critical Low alert within 60 seconds, so that I can provide immediate medical assistance if they are unconscious or impaired."*	8
GT_110	*"As a user, I want the system to bypass Do Not Disturb settings when a Critical Low is detected so that I never miss a life-threatening alert."*	5
GT_111	*"As a user, I want the emergency packet sent to my contacts to include my last known glucose value and trajectory, so my caregiver understands the severity before they reach me."*	3

UC4 for Predictive Alert System: Clinical Threshold Steering

User Stories for UC2-PAS (Subtotal = 18 pts):

ID #	Description	Story Pts.
GT_112	*"As an endocrinologist, I want to adjust the predictive sensitivity thresholds for my patient's overnight window so that I can reduce alarm fatigue caused by non-critical fluctuations while they are sleeping."*	8
GT_113	*"As an endocrinologist, I want to view a standardized AGP report for my patient during a 15-minute appointment so I can make data-driven treatment adjustments quickly."*	5
GT_114	*"As an endocrinologist, I want any threshold changes I make to be immediately committed to the prediction engine so that the patient's alerts reflect my clinical judgment without delay."*	3
GT_115	*"As an endocrinologist, I want all my threshold adjustments logged in the system so there is a clinical audit trail for each patient."*	2

Release 1 NFR Stories

NFR Stories (Subtotal = 18 pts):

ID #	Description	Story Pts.
GT_116	*"As a user, I want all my health data encrypted end-to-end and stored in a HIPAA-compliant database so that my medical information is never exposed."*	8
GT_117	*"As a user, I want to log in using multi-factor authentication so that only I and my authorized providers can access my glucose data."*	5
GT_118	*"As a system, all glucose data must be backed up with 0% data loss tolerance so that no patient reading is ever lost."*	5

Release 2 – Lifestyle & Multimodal Integration

UC3 for Predictive Alert System: Correlate Activity with Prediction

User Stories for UC3-PAS (Subtotal = 23 pts):

ID #	Description	Story Pts.
GT_119	*"As an active user, I want to see how my morning run affects my glucose levels over the next 6 hours so I can adjust my basal insulin rates more accurately."*	8
GT_120	*"As a user, I want the app to detect when I am in a workout and adjust its alert sensitivity, so I don't receive false alarms during exercise-induced glucose dips."*	8
GT_121	*"As a user, I want to grant GlucoTrack permission to sync with Apple Health and Fitbit, so my step count and heart rate are factored into my glucose predictions."*	5
GT_122	*"As a user, I want to see a Workout Mode indicator in the app, so I know the prediction engine has adjusted its sensitivity for my current activity."*	2

Use Case: Log Meal and Calculate Carbs

User Stories (Subtotal = 19 pts):

ID #	Description	Story Pts.
GT_123	*"As a user tired of manual carb-counting, I want to select a meal from a database and have the carbs automatically factored into my insulin calculation to reduce decision fatigue."*	8
GT_124	*"As a user, I want to search a nutritional database of at least 500,000 items so I can find the exact meal I am eating and get an accurate carb count."*	5
GT_125	*"As a user, I want my meal log to be stored alongside my glucose readings so I can see how specific foods affect my levels over time."*	3
GT_126	*"As a caregiver, I want to log meals on behalf of my diabetic child so I can track their nutrition even when they are too young to do it themselves."*	3

Use Case: Meal Finding

User Stories (Subtotal = 16 pts):

ID #	Description	Story Pts.
GT_127	*"As a Type 1 diabetic without my supplies, I want to quickly find nearby meal options that won't spike my glucose so I can manage my levels even when I'm unprepared."*	8
GT_128	*"As a user, I want to filter nearby restaurant options by carb content so I can make a safe meal choice within a specified radius."*	5
GT_129	*"As a user, I want the meal finding service to use my current location automatically, so I don't have to manually enter where I am during a stressful low-glucose situation."*	3

Release 2 NFR Stories

User Stories (Subtotal = 8 pts):

ID #	Description	Story Pts.
GT_130	*"As a user, I want my fitness data to sync within 10 seconds so that my activity state is reflected in my predictions in near real-time."*	5
GT_131	*"As a user, I want the nutritional database to return a carb count within 2 seconds of meal selection, so logging doesn't interrupt my meal."*	3

Release 3 – Platform Maturity

Enhanced AGP Reporting

User Stories (Subtotal = 8 pts):

ID #	Description	Story Pts.
GT_132	*"As an endocrinologist, I want to export a patient's AGP report as a PDF so I can include it in their insurance records or share it with other providers."*	5

GT_133	*"As an endocrinologist, I want the AGP report to cover a configurable time period (14, 30, or 90 days) so I can review trends relevant to the patient's current treatment phase."*	3

Multi-Patient Clinic Dashboard

User Stories (Subtotal = 26 pts):

ID #	Description	Story Pts.
GT_134	*"As an endocrinologist, I want to monitor up to 50 concurrent patient profiles from a single clinic dashboard so I can identify patients who need urgent intervention between appointments."*	13
GT_135	*"As a clinic administrator, I want to manage provider licenses and assign patients to endocrinologists, so the platform scales across our clinic's full patient roster."*	8
GT_136	*"As an endocrinologist, I want to filter my patient list by alert severity so I can prioritize the patients most at risk at any given moment."*	5

Performance & Load Hardening NFR Stories

NFR User Stories (Subtotal = 21 pts):

ID #	Description	Story Pts.
GT_137	*"As a system, 100% of emergency SMS notifications must be dispatched within 3 seconds under a load of 5,000 concurrent triggers so that no critical alert is delayed at scale."*	8
GT_138	*"As a system, the platform must support ≥50 concurrent patient profiles per clinic license without degradation in response time so the B2B2C model is viable for real clinic adoption."*	8
GT_139	*"As a system, all GlucoBridge data normalization must complete within 1 second with zero data loss when switching between CGM brands mid-session so hardware transitions are seamless."*	5

Scenarios

Scenario A: The Proactive Intervention (Predictive Analytics)

User: Sarah, a college student with Type 1 Diabetes.

The Situation: Sarah is studying in the library. Her CGM shows her glucose is at 110 mg/dL and stable, but she hasn't eaten in 4 hours.

The Action: GlucoTracker's predictive engine analyzes her historical "Friday afternoon" trends and predicts a "Hypo" (low blood sugar) event within 45 minutes.

The Outcome: The app sends a "Predictive Alert" suggesting a 15g carb snack. Sarah eats a granola bar, avoiding a dangerous mid-study crash.

Scenario B: The Emergency Support Loop

User: Leo, an elderly user whose glucose drops dangerously low while sleeping.

***The Situation*:** Leo's glucose hits 55 mg/dL. He doesn't wake up to the standard alarm.

The Action: The Emergency Services Support module triggers. It sends an urgent SMS with his GPS location to his "Circle of Care" (his daughter and doctor).

The Outcome: His daughter calls him, wakes him up, and ensures he treats the low, preventing a trip to the ER.

Functional Requirements

- The app will display real-time glucose readings.
- The app will generate alerts for high/low glucose levels.
- The app will accurately predict glucose events for up to 30 minutes in advance.
- The app can be used with multiple CGM devices, doesn't matter the manufacturer.

Non-Functional Requirements

Performance	**Requirement**	**Target**
Performance	Alert delivery time	Within 3 seconds of detection
Performance	Glucose data update	Within 5 seconds
Reliability	Data loss rate	0% (system needs to give accurate data)

Security	Data encryption	Industry-standard end-to-end encryption, comply with HIPAA regulations
Security	Authentication	Multi-factor authentication required
Usability	Data access	Within 2 user interactions
Usability	Critical alert readability	Within 2 seconds
Compatibility	CGM interoperability	support most major CGM devices such as Dexcom, Abbott Libre, and Medtronic.

Acceptance Plan

All functional requirements must pass acceptance testing.

- Performance: The overall application interface should take no longer than 5 seconds to update and display glucose readings. Emergency alerts should be sent, and users should be notified within 3 seconds of detecting the issue so that patients can act immediately.
- Reliability: Tests need to be run to verify all data is accurate and stored without any loss
- Security: All data should use end-to-end encryption and include secure data storage and transmission. The system should also ensure secure authentication and access.

System Tests

Test	Description	Metric	Business Link
Clinical Safety (Functional)	Validate 30-minute predictive accuracy	ML model must achieve >= 90% sensitivity for hypoglycemic events in a sandbox dataset	Justifies 'Predictive Clinical Intelligence' differentiator
Interoperability (Functional)	Swap data source from Dexcom API to Medtronic manual entry mid-session	GlucoBridge must normalize data within < 1 second with zero data loss in the trend graph.	Proves 'Hardware Agnosticity' strategy
Stress & Load (NFR)	Simulate 5,000 concurrent "Emergency Support Loop" triggers	100% of SMS/Push notifications must be dispatched within 3 seconds	Supports the "Emergency Services Support" revenue/value driver

Requirements Analysis for the Predictive Analysis System

For the **Predictive Alert System**, we utilize a **Plan-Driven (Structured)** approach for core safety features and an **Agile** approach for the user interface and reporting modules.

Domain Analysis

The diabetes management domain involves monitoring, interpretation, and response to glucose data in real time. It includes multiple interacting stakeholders, devices, and external systems. The application that will be further analyzed is the Predictive alert system, which utilizes machine learning to develop user glucose level trends to allow them to better prepare for high-stress situations.

The domain centers on the **Continuous Glucose Monitoring (CGM) Lifecycle**. It involves high-frequency data ingestion (every 5 minutes), algorithmic trend projection, and a deterministic escalation protocol. The system must act as an intermediary between raw physiological data and clinical intervention.

Problem Analysis

The main problem addressed by GlucoTrack is the lack of actionable insights in diabetes management systems. Users must manually identify trends and predict risks. Current CGM systems are reactive – they alert users when a threshold is already crossed.

Current solutions also show several key gaps:

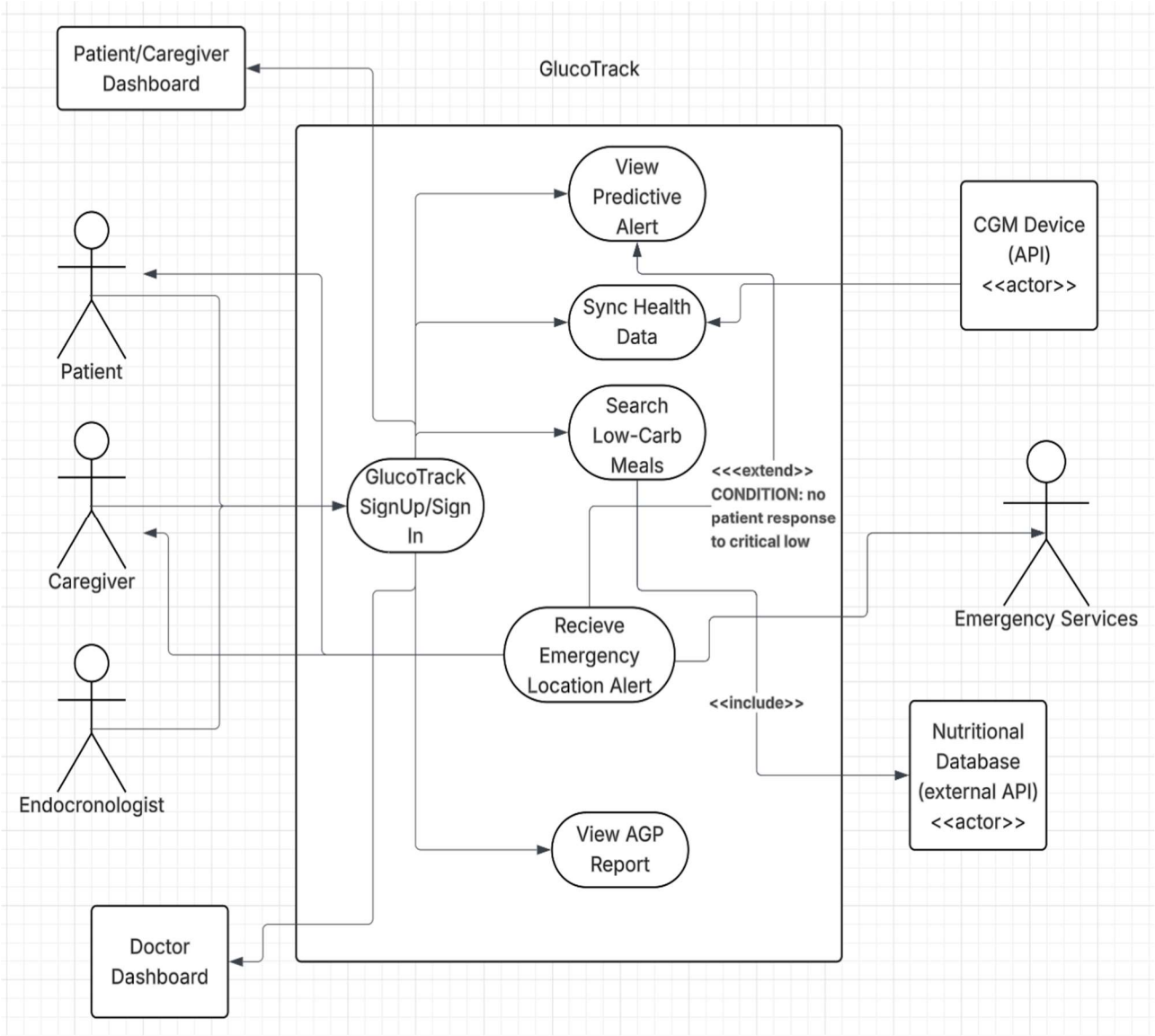
- Limited predictive capabilities
- Poor integration with external health data such as fitness and nutrition information.
- Users due to sleep, exercise, or cognitive fog from low blood sugar cannot respond in time

As a result of these flaws, users face an increased risk of hypoglycemia and hyperglycemia due to delayed or missed warning signs. Additionally, the need to constantly read and interpret complex data contributes to stress, ultimately reducing the user's overall quality of life and making effective diabetes management more difficult.

Solution Analysis

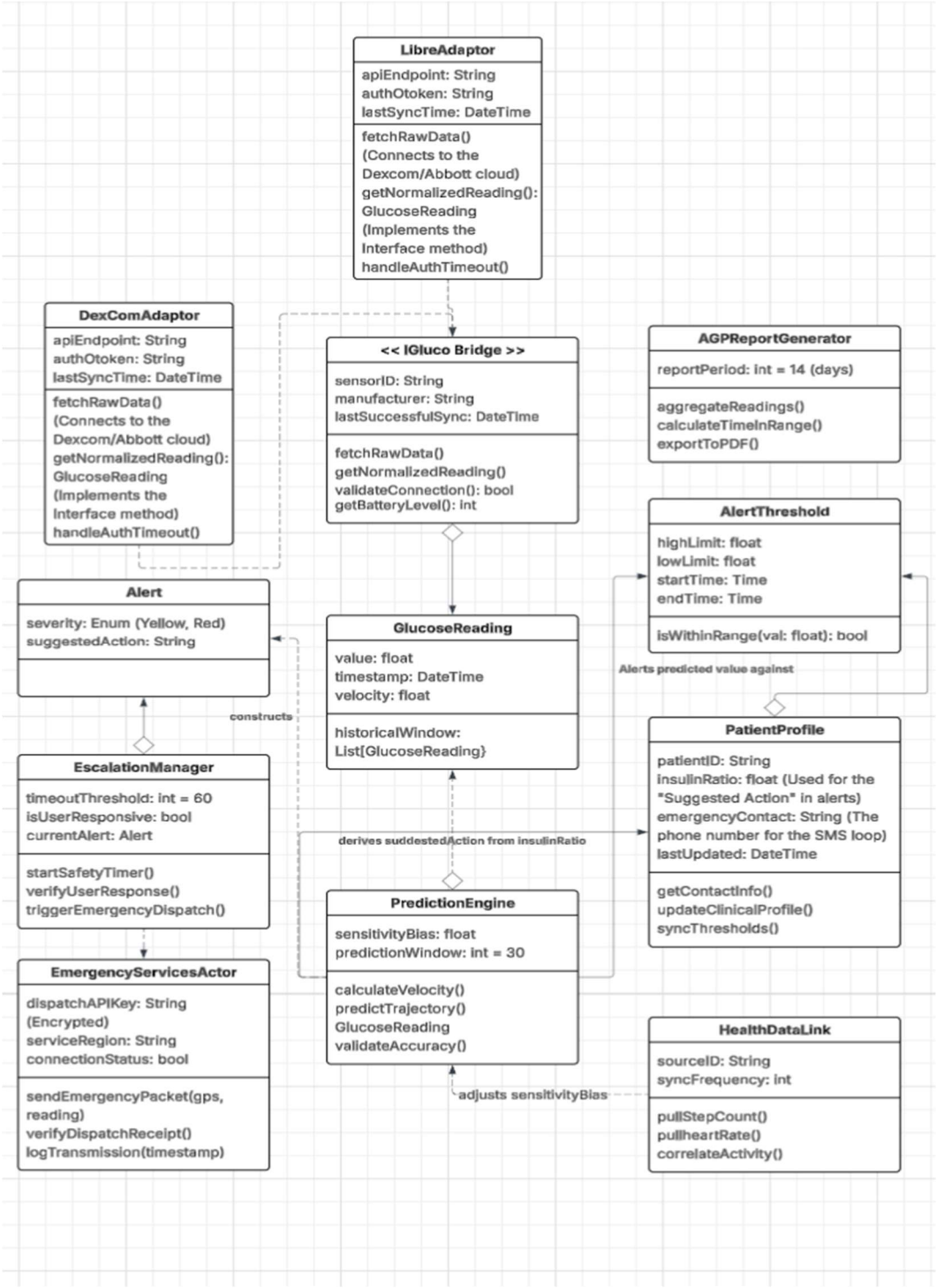
The **Predictive Alert** System application addresses the problem of properly monitoring glucose levels and making proper assessments during stressful situations by showing the user alerts on their mobile device, which are triggered by machine learning (ML) analysis. The predictive analysis application utilizes the CGM device data to identify these trends and alert the user proactively.

To start with, a Use Case Diagram for GlucoTrack that displays the dashboards for both primary and secondary users (as well as external actors).

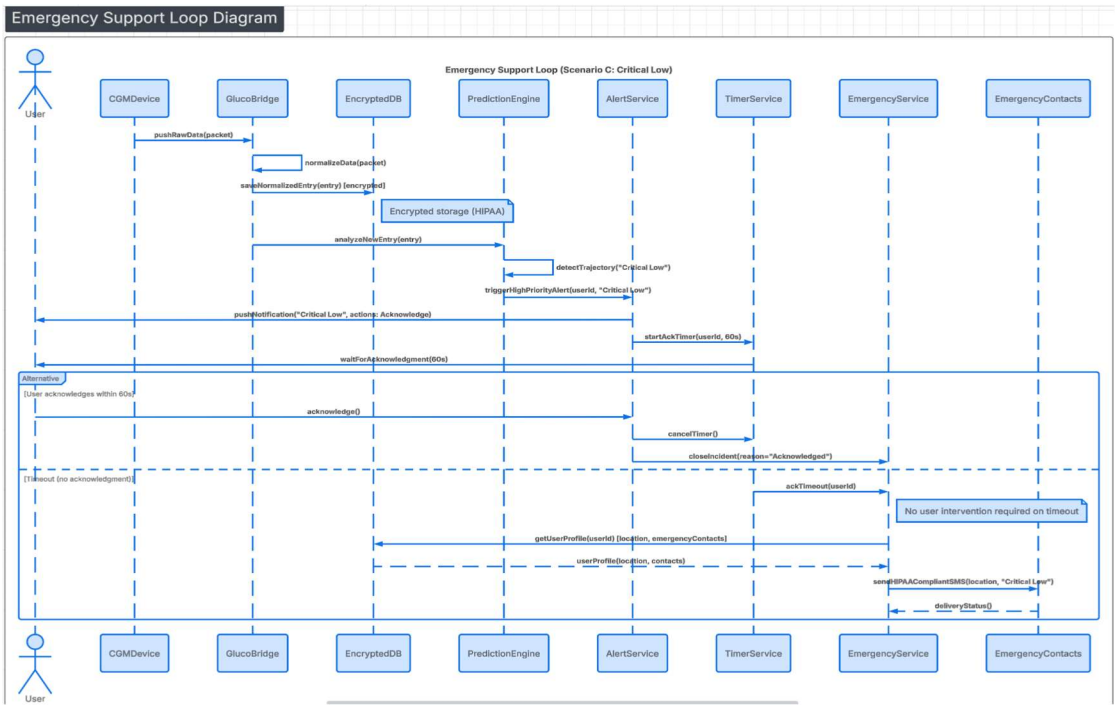


The primary users – patient and caregiver – have a couple of primary actions. Firstly, they must sign into the main app, GlucoTrack, which will house several features for the user (excluding the meal system, additional feature to develop in the future as Priority 2).

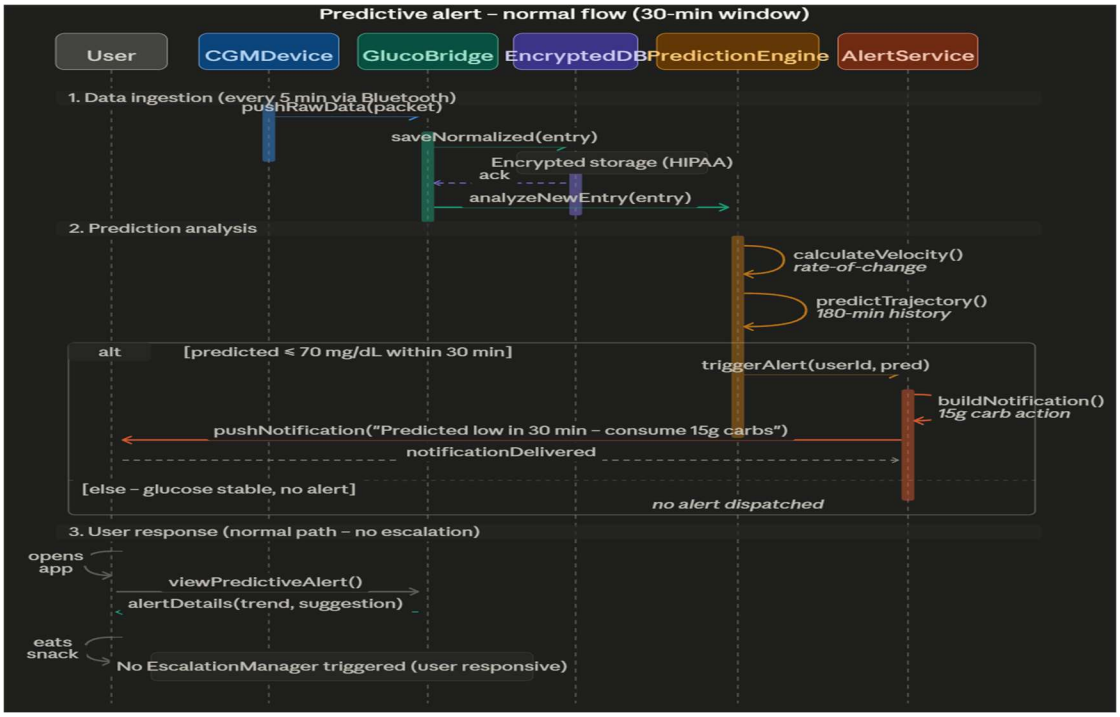
While the Use Case Diagram establishes the functional requirements for the Predictive Alert System, the **DMS Class Diagram** provides the structural solution. It defines the 'Contract' between the GlucoBridge interface and the PredictionEngine, ensuring that hardware-agnostic data is reliably processed into the safety-critical alerts modeled above.



Alert Construction Flow: The *PredictionEngine* retrieves a normalized *GlucoseReading* from *GlucoBridge* and checks the predicted trajectory against AlertThreshold.isWithinRange() to assign severity. It then reads PatientProfile.insulinRatio to compute the suggestedAction. Both values are used to construct an *Alert* object, which is passed to *EscalationManager*. If isUserResponsive remains false after 60 seconds and severity is Red, *EscalationManager* calls triggerEmergencyDispatch() on EmergencyServicesActor.



Below is another sequence diagram that includes the happy path. It makes explicit the three stage internal logic inside the PredictionEngine: calculateVelocity() → predictTrajectory() using 180 minutes of history → threshold check at <= 70 mg/dL. This directly maps to User Story 1's acceptance criteria.



Sub-Features of the Predictive Alert System

AGP Report

The Secondary user, endocrinologist doctor, will be able to sign in and view AGP reports about their patient's glucose monitoring and aid in medical treatments/research, can adjust . Complete with their own dashboard.

Support for Main Feature: The AGP Report (Ambulatory Glucose Profile) is the primary tool the Endocrinologist uses to audit the system's performance. If a patient is experiencing too many "Yellow" alerts at 2 AM, the doctor uses the report to identify the pattern and manually adjust the AlertThresholds. This creates a Human-in-the-Loop safety check.

Sync Health Data

The primary user will use the sync health data interface to connect their CGM to the application as well as other health/activity apps to interact with the predictive analysis in order to find how the user's activities impact their levels and suggest ways to improve their results and help them to understand their symptoms/common levels.

Support for Predictive Alert System: By syncing with fitness and activity apps (Inbound Logistics), the system gains a Multimodal Data Stream. For example, if the system detects a rapid glucose drop during a recorded "High-Intensity Interval Training" session, it can adjust its prediction sensitivity –reducing false positives (alarm fatigue) and improving accuracy of the 30-min window.

Key Processes

- Glucose monitoring
- Predictive analysis of glucose trends
- Alert generation
- Clinical reporting and provider interaction

Constraints

- HIPAA Regulatory requirements
- High reliability and accuracy requirements to ensure personal safety
- Real-time processing expectations

Detailed Use Cases for Predictive Alert System

Use Case 1: Generate Proactive Predictive Alert

Actor: System (ML Engine), User

Description: The system forecasts future glucose levels and proactively alerts the user to prevent a medical emergency.

Steps:

1. The Prediction Engine retrieves the last 3 hours of normalized data from GlucoBridge.
2. The system calculates the glucose velocity (rate of change).
3. The ML model identifies a trajectory that will cross a high or low threshold within 30 minutes.
4. The system generates a "Proactive Notification" with a specific correction suggestion (e.g., "Consume 15g carbs").
5. The user receives and views the alert on their primary device.

Use Case 2: Escalate Unresponsive Critical Alert

Actors: System, User, Caregiver, Emergency Services

***Description*:** A fail-safe loop that triggers external help if a user is physically unable to respond to a critical low.

Steps:

1. The system detects a "Critical Low" trajectory (< 55 mg/dL) and triggers a high-priority alarm.
2. The TimerService initiates a 60-second countdown for user acknowledgment.
3. The system monitors for a user "I'M OK" interaction.
4. If the timer expires without acknowledgment, the EscalationManager retrieves the user's current GPS coordinates.
5. The system dispatches a HIPAA-compliant emergency packet to the Caregiver and local Emergency Services.

Use Case 3: Correlate Activity with Prediction

Actor: Patient, System

Description: The system adjusts predictive sensitivity based on physical exertion data to reduce false alarms.

Steps:

1. The user grants permission to sync fitness data (e.g., Apple Health, Fitbit).
2. The system pulls real-time "Step" or "Heart Rate" data during a glucose monitoring cycle.
3. The Prediction Engine identifies an active workout state.
4. The system adjusts the "Confidence Interval" of the glucose prediction to account for exercise-induced dips.
5. The application displays a "Workout Mode" status to the user to confirm the adjusted monitoring.

Use Case 4: Clinical Threshold Steering

Actor: Endocrinologist, System

Description: A medical professional reviews data and manually adjusts the AI's sensitivity for better clinical outcomes.

Steps:

1. The Endocrinologist logs into the Provider Dashboard.
2. The system generates and displays the AGP Report (Ambulatory Glucose Profile).
3. The doctor identifies patterns of "Alarm Fatigue" (too many false positives) during specific time windows.
4. The doctor inputs new sensitivity parameters into the AlertThreshold settings for that patient.
5. The system updates the Prediction Engine logic to utilize the new professional-steered thresholds.

Detailed User Stories with Acceptance Criteria

Story 1: 30-Minute Predictive Alert

User Story

"As a college student with Type 1 diabetes, I want the system to notify me 30 minutes before my glucose is predicted to hit a 'Low' threshold, so that I can consume carbohydrates and prevent a hypoglycemic episode during high-stress events like exams." (High Priority)

Acceptance Criteria:

- The PredictionEngine must calculate trends using at least 180 minutes of historical GlucoseReading data.
- The alert must trigger if the projected value is <=70 mg/dL within a 30-minute window.
- The notification must suggest a specific action (e.g., "Consume 15g carbs") based on the user's InsulinRatio.

Story 2: Emergency SMS with GPS Location

User Story

"As a parent or caregiver of a diabetic child, I want to receive an emergency SMS with the child's real-time GPS location if they do not acknowledge a 'Critical Low' alert within 60 seconds, so that I can provide immediate medical assistance if they are unconscious or impaired."

Acceptance Criteria:

- The EscalationManager must initiate a TimerService immediately upon a "Critical Low" (< 55 mg/dL) detection.
- If isUserResponsive remains false after 60 seconds, the system must bypass "Do Not Disturb" settings for the caregiver.
- The emergency packet must include the last known glucose value and precise GPS coordinates.

Story 3: Overnight Threshold Adjustment (Endocronologist)

User Story

"As an endocrinologist, I want to adjust the predictive sensitivity thresholds for my patient's overnight window, so that I can reduce 'alarm fatigue' caused by non-critical fluctuations while they are sleeping"

Acceptance Criteria:

- The Provider Dashboard must allow the doctor to set time-specific AlertThresholds (e.g., 12 AM – 6 AM).
- Changes made by the doctor must be immediately "committed" to the PredictionEngine logic for that specific user.
- The system must log these adjustments in the AlertTable for clinical audit purposes.

Section 4: Architecture

Subsystem Model

Data Ingestion (GlucoBridge)	Responsible for “Inbound Logistics." It manages Bluetooth handshakes and API polling for Dexcom/Libre/Medtronic and normalizes the data.
Intelligence Subsystem	Houses the ML models and the Prediction Engine. It runs independently of the UI to ensure 24/7 background monitoring.
Notification & Emergency Subsystem	Handles the "Outbound Logistics." This subsystem manages escalating alert logic and secure SMS dispatching.
User Engagement Subsystem	Manages the React.ts frontend, the Meal Finding Service, and health/fitness integrations.

Target Environment

Component	Technology	Justification
Mobile Client	React Native / TypeScript	Provides a Responsive UX across iOS and Android while maintaining strict type safety for medical data.
Application Server	FastAPI (Python 13.2+)	High-performance asynchronous framework. Native compatibility with ML libraries allows for zero-latency predictions.
Cloud Hosting	AWS (Amazon HealthLake & Lambda)	HIPAA-eligible infrastructure. Lambda handles isolated GlucoBridge connectors for different CGM brands.

Database	PostgreSQL (RDS)	Supports structured relational data for glucose trends with robust encryption-at-rest.
ML Runtime	AWS SageMaker	Allows for the "AI Refinement" mentioned in our Tech Support activities to scale as our anonymized dataset grows.

System Architecture

We utilize a 3-Layered Web Architecture optimized for real-time medical monitoring.

Layer 1: Presentation (Mobile Client)

The user only sees what matters. Instead of overwhelming graphs, the mobile layer uses User-Centered Design to show a simple "Traffic Light" status (Green = Stable, Yellow = Warning, Red = Emergency).

Layer 2: Service (FastAPI Micro-Kernel)

This is the "Logic Layer" where the GlucoBridge resides.

- Middleware: When a Dexcom sensor sends data, the Bridge "normalizes" it so the rest of the system doesn't need to know what brand of hardware is being used (Hardware Agnosticism).
- Authentication: Ensuring only the user and their doctor can see the data (HIPAA Compliance).

Layer 3: Persistence & Analytics (The Vault)

This is our secure backend on AWS:

- The Prediction Engine: Because we use FastAPI, the AI can analyze the "Vault" data instantly to provide the 30-minute forward-looking window.
- The Provider Dashboard: A separate entry point for doctors to view patient trends without needing to access the mobile app directly.

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